

NYC 311 Service Request Closure Prediction

Predicting Whether a Service Request Is Closed Within 24 Hours

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1 Executive Summary

This report documents a machine learning solution designed to predict whether an NYC 311 service request will be closed within 24 hours of its creation. The final reproducible solution uses a `CatBoostClassifier` trained from 16 selected raw input columns. After preprocessing and feature engineering, these inputs become 53 final model features, 35 of which are categorical.

The project is a binary classification task on tabular civic-service data. Controlling target leakage is the most important methodological constraint: `Closed Date` is required to construct the target, but it is unavailable at inference time and is therefore excluded from the model inputs.

The final saved run reports a training accuracy of 0.9181 and a validation accuracy of 0.9032, with a train-validation gap of 0.0148. Five-fold cross-validation gives a mean validation accuracy of 0.8997 with a standard deviation of 0.0018, suggesting stable generalization across stratified folds. Since the official project metric is accuracy, this is the main evaluation criterion used in the report.

2 Problem, Data Understanding, and Exploratory Data Analysis

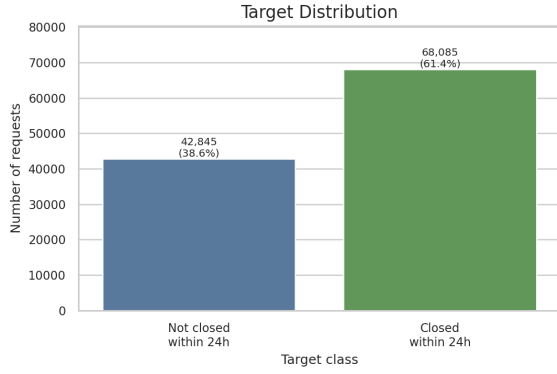
The objective is to predict whether an NYC 311 service request will be closed within 24 hours of being opened. Each row represents one service request and contains information about the issue type, agency, location, routing channel, and timestamps.

The target variable is defined as:

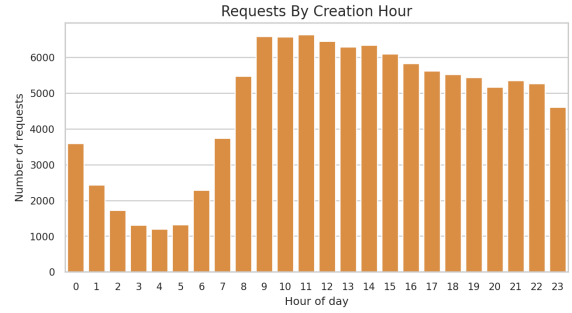
$$y = \begin{cases} 1, & \text{if the request is closed within 24 hours} \\ 0, & \text{otherwise} \end{cases}$$

More precisely, class 1 is assigned only when the elapsed time between `Created Date` and `Closed Date` is non-negative and less than or equal to 24 hours. Requests not observed as closed within that window, including rows with missing or invalid `Closed Date`, are assigned class 0. This definition assumes that the dataset represents historical requests for which the closure outcome can be observed; if very recent still-open requests were present, assigning missing `Closed Date` values to class 0 could introduce label noise due to right-censoring.

The constructed target is moderately imbalanced. About 61.38% of training observations are closed within 24 hours, while 38.62% are not. Therefore, accuracy must be interpreted relative to a naive majority-class baseline of about 61.38%.



(a) Target distribution.



(b) Requests by creation hour.

A key methodological choice is to exclude **Closed Date** from the model inputs. Including it would introduce target leakage, because it contains information available only after the request has been closed. The final script uses 16 selected raw input columns covering request creation time, agency, complaint type, geographic information, and request-routing context.

The exploratory analysis highlighted three main patterns. First, closure behavior varies across time, making hour-of-day and day-of-week features useful. Second, strong heterogeneity appears across categories such as agency, complaint type, and borough, suggesting that categorical interactions are predictive. Third, several retained variables contain meaningful missingness: **Additional Details** is missing in about 63.01% of requests, **Location Type** in 15.55%, **City** in 5.19%, and latitude/longitude in 3.57%.

3 Preprocessing

The target is created from **Created Date** and **Closed Date**. Both timestamps are parsed using the project date format, and invalid values are coerced to missing timestamps. The elapsed time in hours is computed as:

$$\text{hours_to_close} = \frac{\text{total_seconds}(\text{Closed Date} - \text{Created Date})}{3600}$$

Several numeric columns contain formatting issues such as extra spaces, non-breaking spaces, or commas. These fields are cleaned before numeric conversion. Latitude and longitude receive special handling because commas may appear as decimal separators.

The final reproducible pipeline uses a fixed selected-column configuration rather than dynamically removing columns based on missingness. Categorical missing values are filled with the visible token **UNKNOWN**, allowing the model to learn whether missingness is informative. Numeric values that cannot be converted are left as missing values, which CatBoost can handle directly.

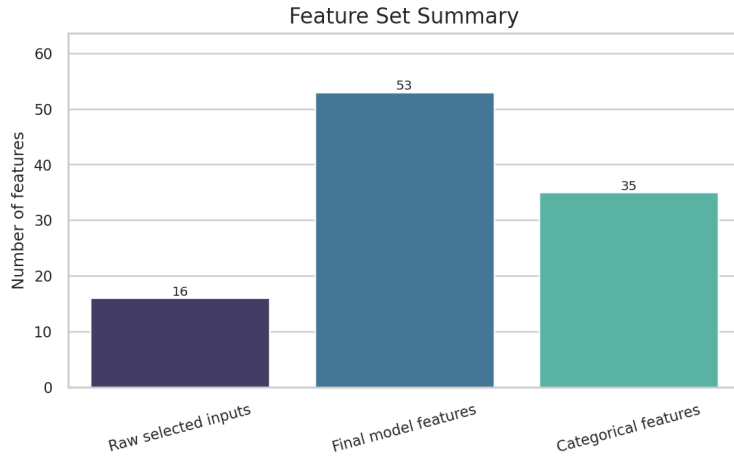
4 Feature Engineering

Feature engineering is the core contribution of the project. **Created Date** is not kept as a raw feature after temporal extraction. Instead, the pipeline creates hour, day-of-week, day-of-month, weekend, business-hour, minute-of-day, and cyclical sine/cosine encodings for hour and weekday. These temporal features are available at request creation time and therefore do not introduce target leakage.

The pipeline also enriches numeric and geographic information. It converts selected numeric fields into usable numeric form, creates categorical versions of location-like numeric fields, and computes a simple distance-from-NYC-center feature using latitude and longitude.

The most important engineered variables are categorical interactions. These include pairwise and higher-order combinations involving agency, complaint type, descriptor, borough, zip category, location type, community board, police precinct, and open data channel type. These interaction features allow CatBoost to capture operational patterns that depend on combinations of request attributes.

The final feature matrix contains 53 features: 15 retained raw inputs excluding **Created Date**, 10 temporal features, 4 numeric/geographic enrichment features, 12 pairwise categorical interactions, 8 higher-order categorical interactions, and 4 missingness/length features.



5 Exploratory Feature Screening and Baselines

The notebooks were used to explore preprocessing alternatives, feature screening, and baseline models. One notebook examines near-constant variables, highly correlated numeric features, and Random Forest feature importance. Another compares a logistic regression baseline with stronger tabular classifiers.

These experiments supported the final choice of CatBoost, but they should be interpreted as exploratory rather than as the authoritative final evaluation. The official saved evaluation is the final CatBoost run reported by the repository training script.

6 Model Choice

Logistic regression provides a simple linear baseline, while gradient boosting models test whether non-linear interactions improve performance. This comparison is useful because the dataset contains many categorical variables and likely depends on interactions between agency, complaint type, descriptor, location, and channel.

The final reproducible solution uses `CatBoostClassifier`. This choice is motivated by four considerations. First, CatBoost handles categorical variables natively. Second, it accommodates missing numeric values, reducing the need for aggressive imputation. Third, it is well suited to tabular problems where nonlinear interactions matter. Fourth, it keeps the pipeline compact because raw categorical variables, engineered categorical interactions, and numeric features can all be used within a single model framework.

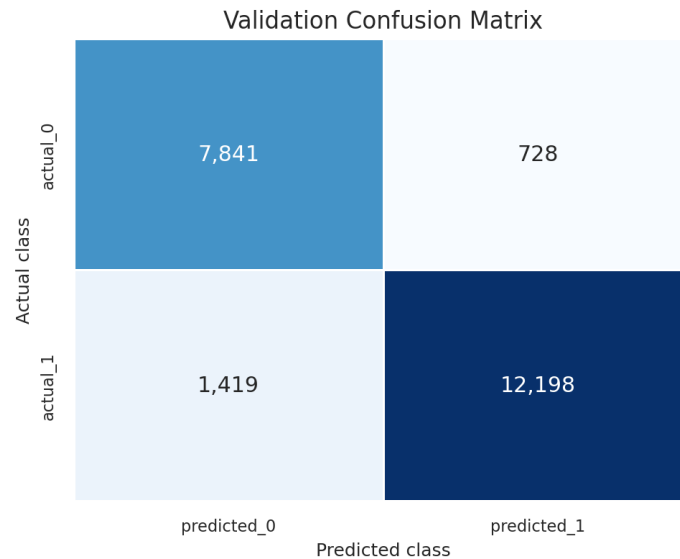
7 Results and Evaluation

According to the saved final run, the CatBoost pipeline achieves a training accuracy of 0.9181 and a validation accuracy of 0.9032. The train-validation gap is 0.0148, suggesting good generalization under the random stratified validation scheme.

Since the majority class accounts for about 61.38% of the training set, a naive classifier that always predicts the most frequent class would obtain roughly 61.38% accuracy. The final model improves on that baseline by about 28.94 percentage points.

Five-fold stratified cross-validation gives a mean validation accuracy of 0.8997, with a validation standard deviation of 0.0018 and a mean train-validation gap of 0.0168. This indicates that performance is stable across folds.

The final script also saves a validation confusion matrix and classification report. On the validation set, the positive class achieves precision 0.9437, recall 0.8958, and F1-score 0.9191.



Because validation is based on random stratified splits, the reported performance should be interpreted as an estimate of in-distribution generalization rather than as a full simulation of future deployment under temporal drift.

Remaining errors are plausible because actual closure time may depend on information unavailable at request creation time, such as workload, staffing constraints, weather, public holidays, or temporary disruptions.

8 Conclusion

This project demonstrates that the likelihood of an NYC 311 request being closed within 24 hours can be predicted with high accuracy using information available at creation time. The effectiveness of the solution is driven by a leakage-aware target definition, temporal and geographic feature engineering, categorical interaction features, and CatBoost.

A key insight is that much of the predictive signal lies in structured metadata, especially interactions between agency, complaint type, descriptor, location, and channel. This explains why a compact pipeline can achieve strong validation performance.

The model also has limitations. It relies only on available structured data and cannot account for external factors such as weather, holidays, workload changes, or staffing constraints. Textual fields are used mainly through raw categorical values and simple missingness or length indicators, without deeper semantic representation. Finally, the validation strategy is based on random stratified splits rather than a time-based split, which may underestimate temporal drift.

Future improvements could include richer text representations, systematic hyperparameter optimization, time-aware validation, and external contextual features. Overall, the proposed approach provides a reproducible solution that captures the main drivers of 24-hour request closure.